

Transformer Loading Analysis - TGI Use Case Notes

[UC26] - Awesense Platform Use Case



Overview

This manual is intended to:

- Provide instructions to create the selected use case in True Grid Intelligence (TGI).

For more details about Transformer Loading Analysis and how it can be analyzed using Awesense Platform, please refer to the [UC26 - Transformer Loading Analysis.pdf](#) document.

TGI Use Case Notes

This document shows the implementation process of the mentioned use case in the TGI visualization interface, along with a description of how to create such a use case on your data.

The transformer loading analysis can be found directly in the TGI for a transformer element if that transformer has measured data¹. The easiest way is to find a transformer via the TGI Explorer in the main menu (select from the main menu - top hamburger icon) and then select the transformer from the map or table. Clicking on a transformer will open a sidebar on the left side with additional information, and at the bottom of this sidebar is the "Phase Balance" section where clicking on "View Report" will display this report.

The screenshot displays the True Grid Intelligence (TGI) interface. On the left sidebar, the "Grid Element" section shows coordinates (38.723687, -82.339262) and a "Zoom to Grid Element" button. Below this, the "Trace" section has radio buttons for Source, Down, Same Voltage, Connected, Nearby One Way, Nearby Other Way, Nearby Both Ways, and All Sources. The "Phase Balance" section shows a "Past week" filter and a "View Report" button highlighted with a red box. The "Grid Element Details" section shows fields for label (17AY23-8-10), my_feeder (0704), name (17AY23-8-8-10), and rating_kva (15). The main area features a map with three transformer locations labeled 17AY23-8-9, 17AY23-8-10, and 17AY23-8-11. Below the map is a table with 15,800 transformers. The table has columns: ID, Circuit, Element Type, Phases, Latitude, Longitude, Is Producer, and Is Consumer. The table is sorted by ID, and the first few rows are visible.

ID	Circuit	Element Type	Phases	Latitude	Longitude	Is Producer	Is Consumer
10001	North Central Zone	Transformer	C	39.254601464	-82.533343845	No	No
10005	North Central Zone	Transformer	A	38.596587474	-82.431991662	No	No
10007	North Central Zone	Transformer	B	39.180105578	-82.055933487	No	No
1001	North Central Zone	Transformer	B	38.723686737	-82.339261635	No	No
10010	North Central Zone	Transformer	A	38.8827073	-82.692737045	No	No
10012	North Central Zone	Transformer	A	38.816414542	-82.324232864	No	No
10013	North Central Zone	Transformer	A	38.834501148	-82.518109731	No	No
10014	North Central Zone	Transformer	B	38.769297398	-82.764101722	No	No
100145	North Central Zone	Transformer	C	38.600791303	-82.415557345	No	No
10017	North Central Zone	Transformer	A	38.668167494	-82.411648443	No	No

¹ Transformer measured data can be originating directly from measurement device installed on given transformer or TGI has option to aggregate downstream consumption and infer the load flow on the transformer this way.

The basic interface of the Transformer Technical Analysis report offers information about the analyzed transformer and its attributes. Further, the interface offers quick information about the percentage of the time the transformer is under or over-loaded.

Transformer Technical Analysis

Transformer Overview

Element ID:	transformer_92
commission_date:	1995-01-01
installation_type:	station
is_operation_status:	Yes
label:	tr_92
latitude:	49.26829145
load_loss:	1117
longitude:	-123.10273
maintenance_type:	dry
name:	tr_92
no_load_loss:	194
ownership:	utility
primary_voltage:	12500
rating_kva:	15
secondary_voltage:	400
voltage_level:	MV/LV

First data point - 18 May, 2023 12:56 PM

PF Override: PF

Percentage of Time Overloaded

35.8%

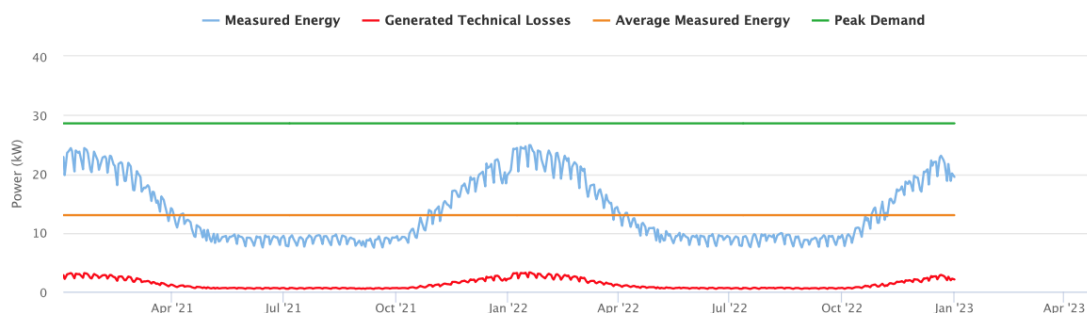


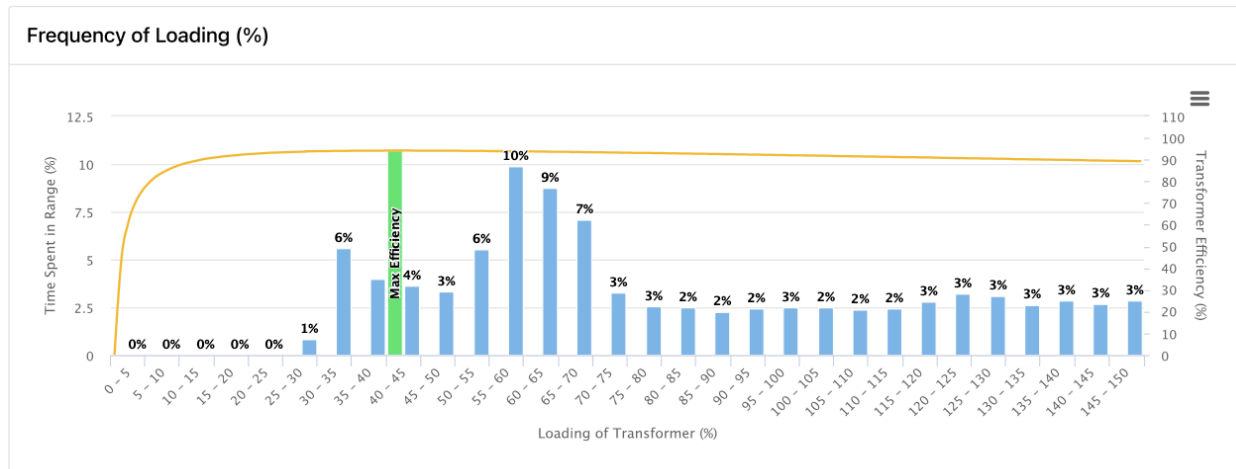
Percentage of Time Underloaded

0.845%



Transformer Performance





The interface also offers the possibility to choose different time periods.

The Transformer Performance graph shows the transformer load pattern over the selected period and also shows other indicators such as Peak Demand, Generated Technical Losses and Average Measured Energy.

The last graph shows a histogram of the transformer loading. Here you can clearly see how the transformer was loaded most often. The transformer loading efficiency (yellow) is also shown along with the maximum efficiency loading ratio (green column).

General Note

- This description was created for TGI version 3.10.0. For future versions of TGI, the process might be slightly different.
- Access to the different parts of TGI varies depending on the permissions of the user.
- Users on the Sandbox environments currently have read-only permissions, therefore, this use case may be accessible only for viewing, not creation or editing. If you are interested in the latter, please contact Awesense.
- Some of the data for this use case may be available only on Sandbox tiers 2 & 3, but not on tier 1. If you are interested in upgrading your tier access, please contact Awesense.
- For more information please contact tools@awesense.com.